Lecture 1b: Utility functions quantify happiness and are used to rank order competing alternatives

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Objectives: In this lecture we will:

- **O1**: Introduce the basic premise of utility functions and rational choices
- **O2**: What is the mathematical form and behavior of some common utility functions?
- **O3**: Introduce the concepts of marginal utility, discrete choice models, and their applications (material for next time in L2a)

Lecture L1b notes and examples: https://bit.ly/45jc70t

An individual decision-maker (agent) is given a set of n objects or features, $X = \{x_1 \ldots, x_n\}$. A utility function ranks the agent's preference for combinations of these objects or features:

$$U(x_1, x_2, \ldots, x_n) = u \tag{1}$$

where u is a real number called the `utility` which has units of `utils`.

- The utility function $U : X \to \mathbb{R}$ is unique only up to an order-preserving transformation.
- Utility functions are `ordinal`, i.e., they rank-order bundles but do not measure differences between bundles.


HARD CANDY
WATERMELON

$$= x_1$$


GREEN APPLE
NATURALLY &
ARTIFICIALLY FLAVORED

$$= x_2$$


HARD CANDY
BLUE RASPBERRY
ARTIFICIALLY FLAVORED

$$= x_3$$


CHERRY
ARTIFICIALLY FLAVORED

$$= x_4$$


HARD CANDY
GRAPE
NATURALLY &
ARTIFICIALLY FLAVORED

$$= x_5$$

Rational choice theory

Rational choice theory is founded on the idea that individuals make decisions that maximize their utility. The theory assumes that individuals always make prudent and logical decisions that provide them with the highest personal utility. In other words, when faced with competing alternatives, a decision-making agent will *always* choose the option that maximizes their utility.

The utility function $U : X \rightarrow \mathbb{R}$ can be used to rank order the preference for different choices. For example, suppose we have two bundles of goods, services, or states option(s) $A \in X$ and $B \in X$ that we must decide between:

- If the decision maker *strictly prefers* option A over B, denoted as $A \succ B$, the utility $U(A)$ is always greater than $U(B)$, i.e., $U(A) > U(B)$.
- If the decision maker is *indifferent* between option A and B, denoted as $A \sim B$, then the utility $U(A) = U(B)$.
- If the decision maker *weakly prefers or is indifferent* between option A over B, denoted as $A \succsim B$, then the utility $U(A) \geq U(B)$.

Key Idea of Rational Choice Theory

When faced with competing alternatives, i.e., different options A and B, a rational decision maker will *always* choose the option that maximizes their utility.

$$x_1 \succsim x_2 \qquad x_2 \succ x_3 \qquad x_3 \succ x_4 \qquad x_4 \succ x_5$$

$$U(x_1) \succsim U(x_2) \quad U(x_2) \succ U(x_3) \quad U(x_3) \succ U(x_4) \quad U(x_4) \succ U(x_5)$$

For these flavors, I have a *strict preference*

For these flavors, I *weakly prefer or are indifferent to* watermelon versus green apple.

O2: What is the mathematical form and behavior of some common utility functions?

A `linear` utility function assumes an individual's utility is directly proportional to the feature variables, e.g., the quantity of a good or service consumed, the state of the world, etc. A `linear` utility function is given by:

$$U(x) = \alpha^{T} \cdot x = \sum_{i=1}^{n} \alpha_{i} x_{i} \tag{2}$$

The term $\alpha^{T} \cdot x$ denotes the inner (or scalar) product between the $n \times 1$ parameter vector α and the $n \times 1$ vector of feature variables x, where $x_i \geq 0$ and $\alpha_i \geq 0$.

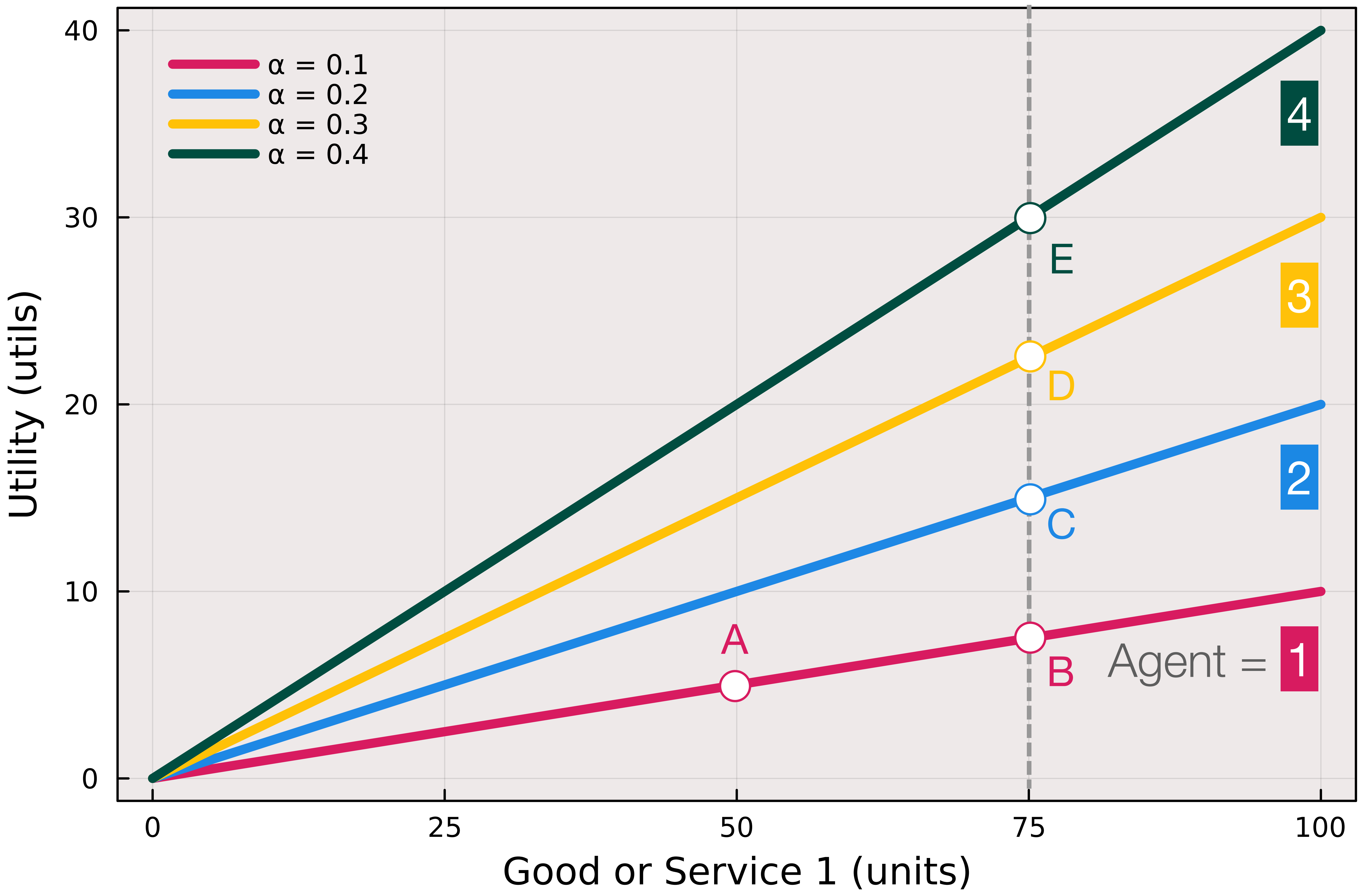
Utility: Linear

$U_1(A) = U_1(50\times$  $)$

$U_1(B) = U_1(75\times$  $)$

$$U_i(75) > U_i(50)\ \forall i$$

More is always better (for all agents)



A `logarithmic` utility function assumes that an individual's utility is a function of the logarithm of the weighted sum of the feature variables, e.g., the quantity of a good or service consumed, the state of the world, etc., plus a constant. A `logarithmic` utility function has the form:

$$U(x) = \ln(\alpha^T \cdot x + \beta) \tag{3}$$

where $\beta > 0$ is a positive (scalar) constant. The term $\alpha^T \cdot x$ denotes the inner (or scalar) product between the $n \times 1$ parameter vector α and the $n \times 1$ vector of feature variables x, where $x_i \geq 0$ and $\alpha_i \geq 0$.

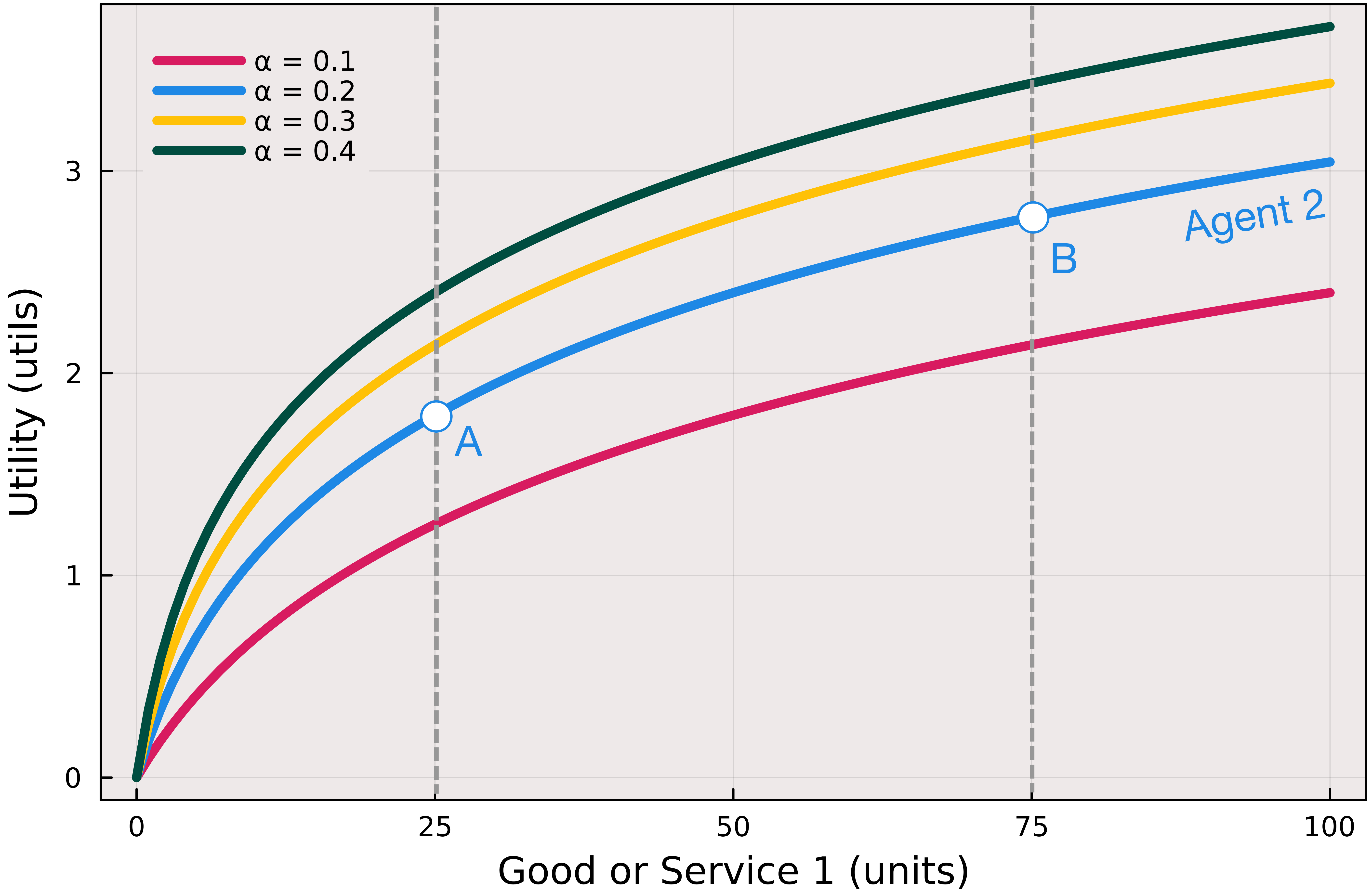
Utility: Log

$U_2(A) = U_2(25\times$  $)$

$U_2(B) = U_2(75\times$  $)$

$$U_i(75) > U_i(25)\ \forall i$$

More is always better (for all agents)



The `Cobb-Douglas` utility function is the product of the n feature variables, thus, it models situations where the features (consumption of goods or services, traits like make, model and color of a car, etc) occur simultaneously. Each feature variable (or good, or service) is raised to a non-negative exponent:

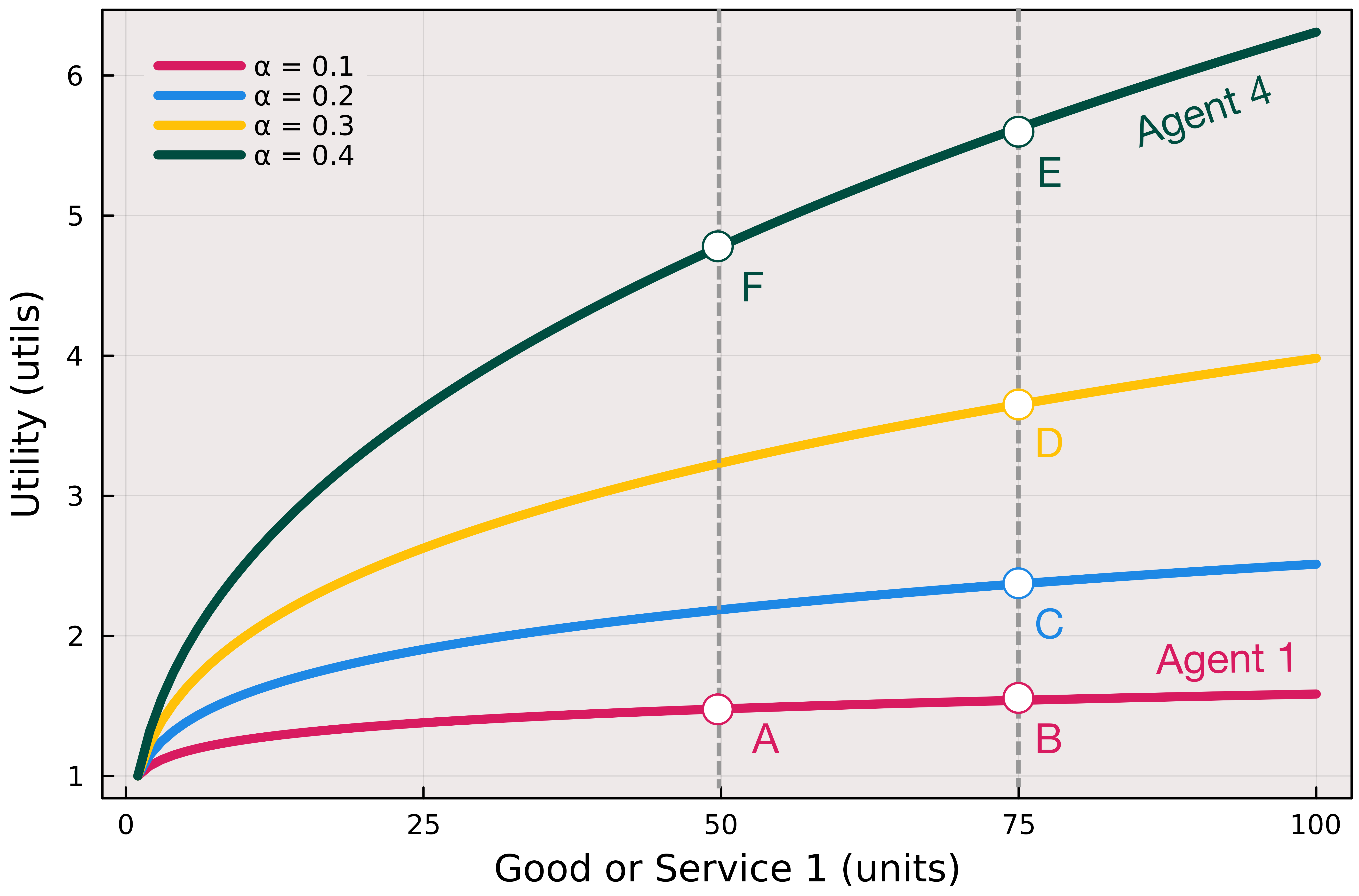
$$U(x_1, \ldots, x_n) = \prod_{i \in 1 \ldots n} x_i^{\alpha_i} \tag{4}$$

In our realization of the `Cobb-Douglas` utility, the exponents must sum to unity $\sum_{i \in 1 \ldots n} \alpha_i = 1$, $x_i \geq 0$, and $\alpha_i \geq 0$.

Utility: Cobb-Douglas

$$U_4(E) > U_4(F)$$
$$U_1(B) \geq U_1(A)$$

More is always better (for all agents) but for Agent 1, it is more subtle (weak)



The `Leontief` utility function, first developed by Wassily Leontief who was later awared a Nobel Prize in Economics for his work, computes the utility for `complementary` states (or configurations) of the world, where each feature variable is scaled by a non-negative constant(s) α. The `Leontief` utility function is given by:

$$U(x_1, \ldots, x_n) = \min \left\{ \frac{x_1}{\alpha_1}, \ldots, \frac{x_n}{\alpha_n} \right\} \tag{5}$$

where $\alpha_i > 0$ and $x_i \geq 0$ for all $i \in 1 \ldots n$.

Utility: Leontief

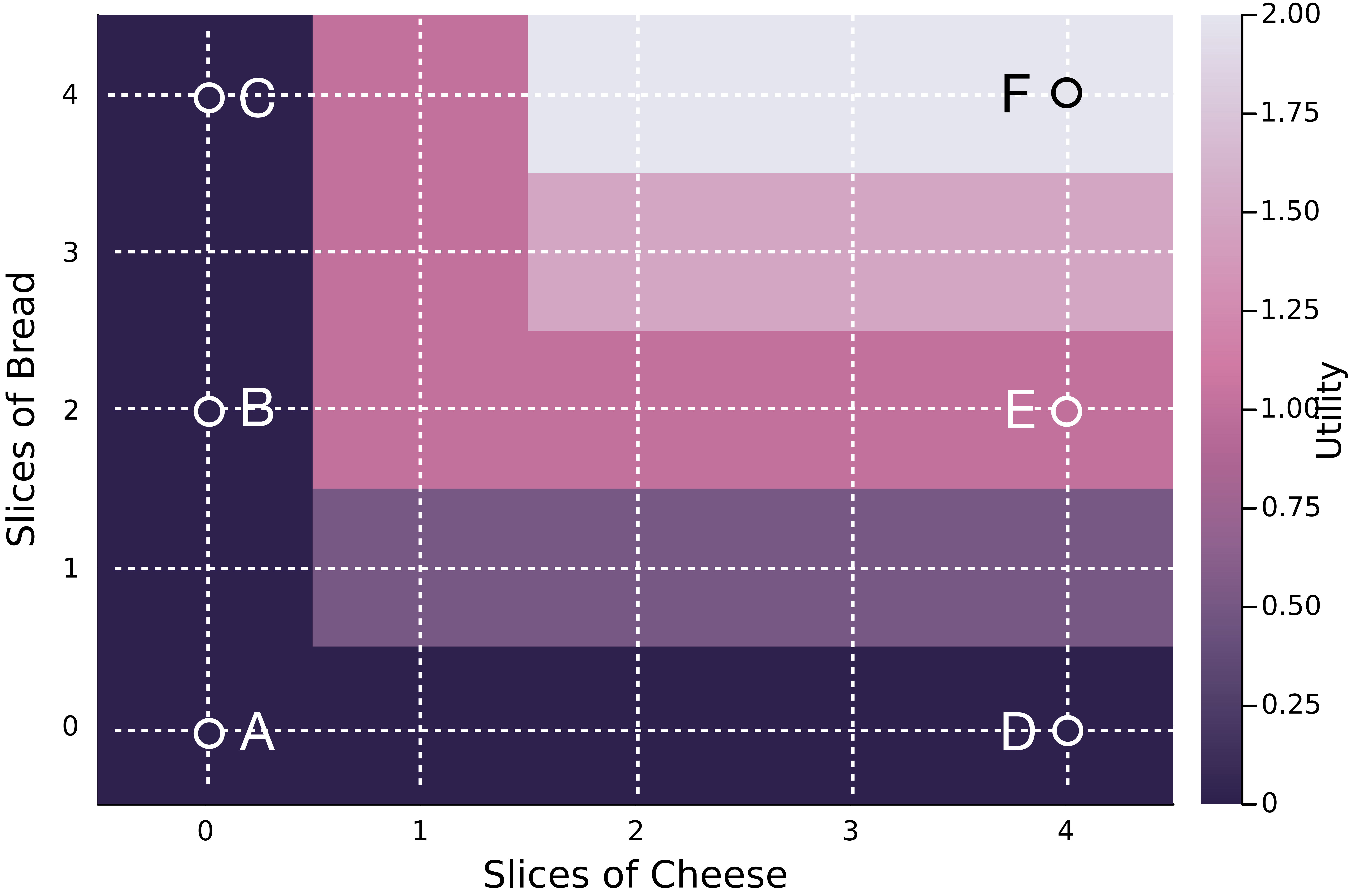
$$U(\text{Bread}, \text{Cheese}) = \min\left(\frac{\text{Bread}}{2}, \frac{\text{Cheese}}{1}\right)$$

No sandwiches

$$U_1(A) = 0$$
$$U_1(B) = 0$$
$$U_1(C) = 0$$
$$U_1(D) = 0$$

Sandwiches

$$U_1(E) = 1$$
$$U_1(F) = 2$$



O3: Introduce the concepts of marginal utility, discrete choice models, and their applications (material for next time in L2a)

Let x_i (for $i \in 1 \dots n$) be a feature, e.g., a good, service or a state of the world which is a member of the set X. Further, let $U : X \to \mathbb{R}$ be a utility function over the set X. The marginal utility of $x \in X$ is the partial derivative of U with respect to x:

$$\bar{U}_{x_i} \equiv \left(\frac{\partial U}{\partial x_i} \right) \tag{6}$$

The marginal utility $\bar{U}_{x_i}$ measures the change in the utility U resulting from small changes in the feature variables x_i.

Discrete choice models assume that people make decisions based on factors that are known to the decision maker, but perhaps unknown to an observer. Thus, there are observable features of the decision, and there are unobservable features that influence the decision that are not modeled. Utility function in this type of situation take the form:

$$U_{ij} = V_{ij} + \epsilon_{ij} \tag{7}$$

where U_{ij} is the utility of alternative j for individual i, V_{ij} is the deterministic component of the utility, i.e., the utility associated with the observable features, and ϵ_{ij} is the random component of the utility.

Today?